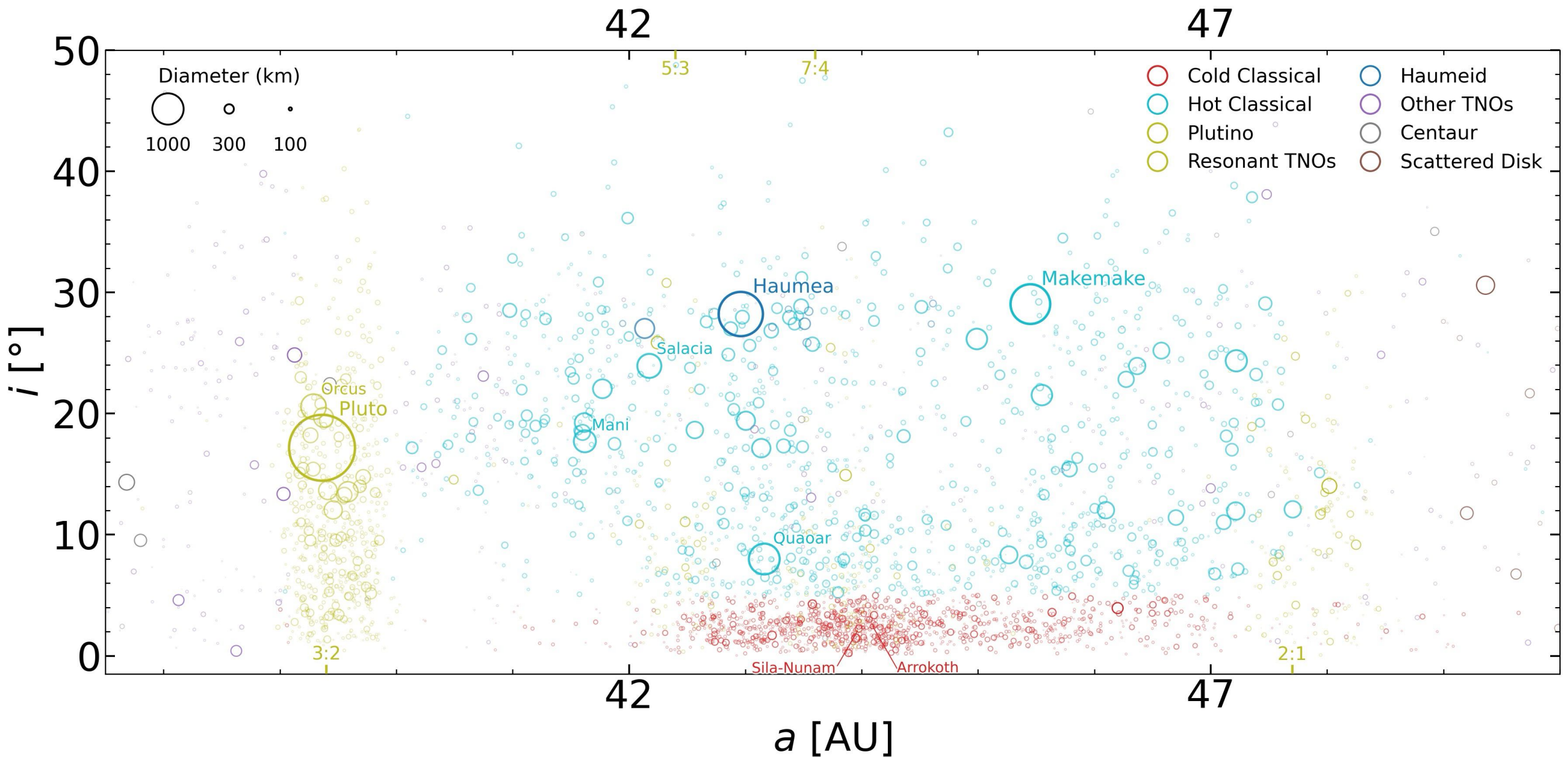


An Origin Story for the Cold Classical Kuiper Belt

Rixin Li

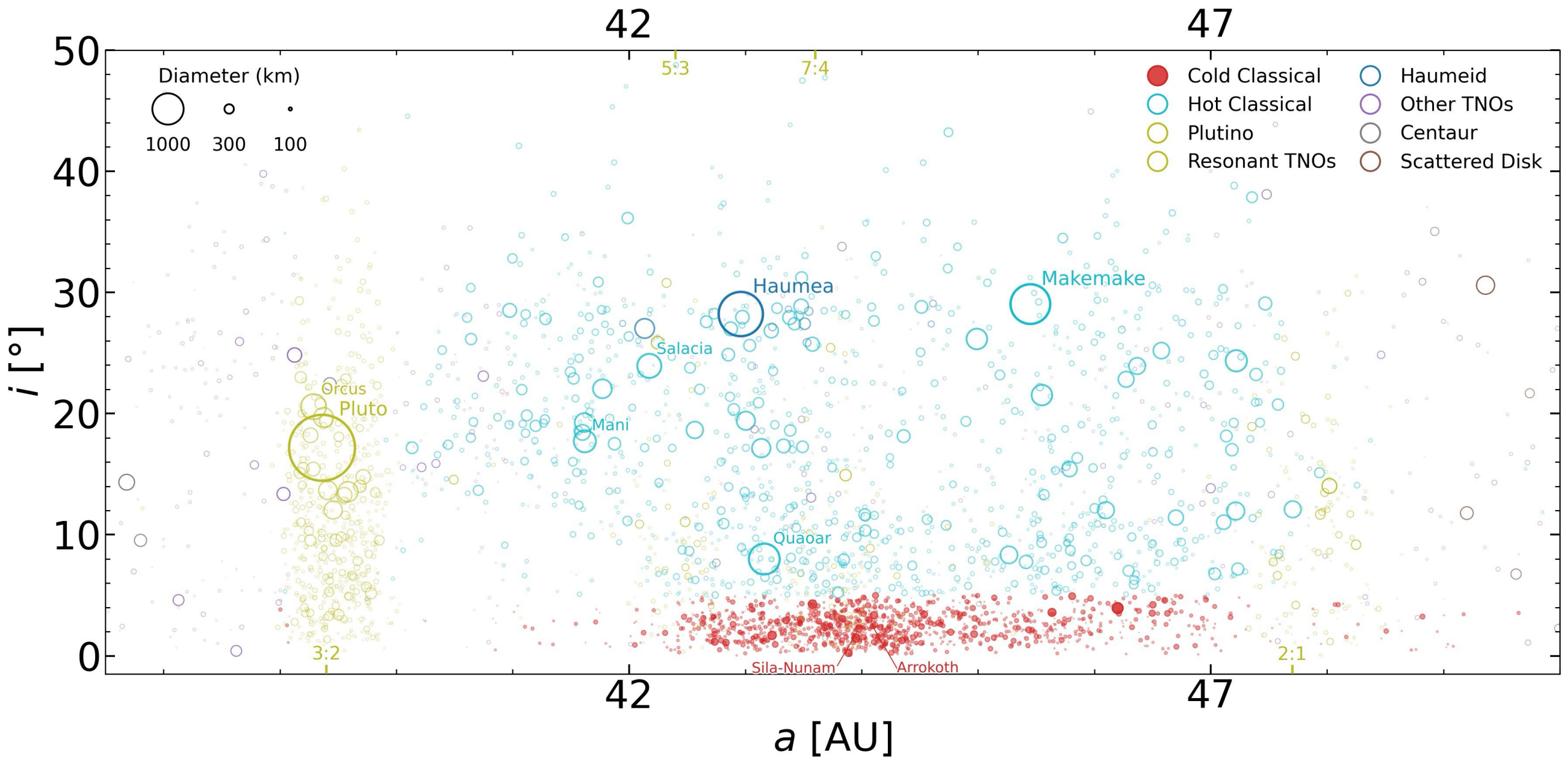
51 Pegasi b Fellow
UC Berkeley

The Kuiper Belt



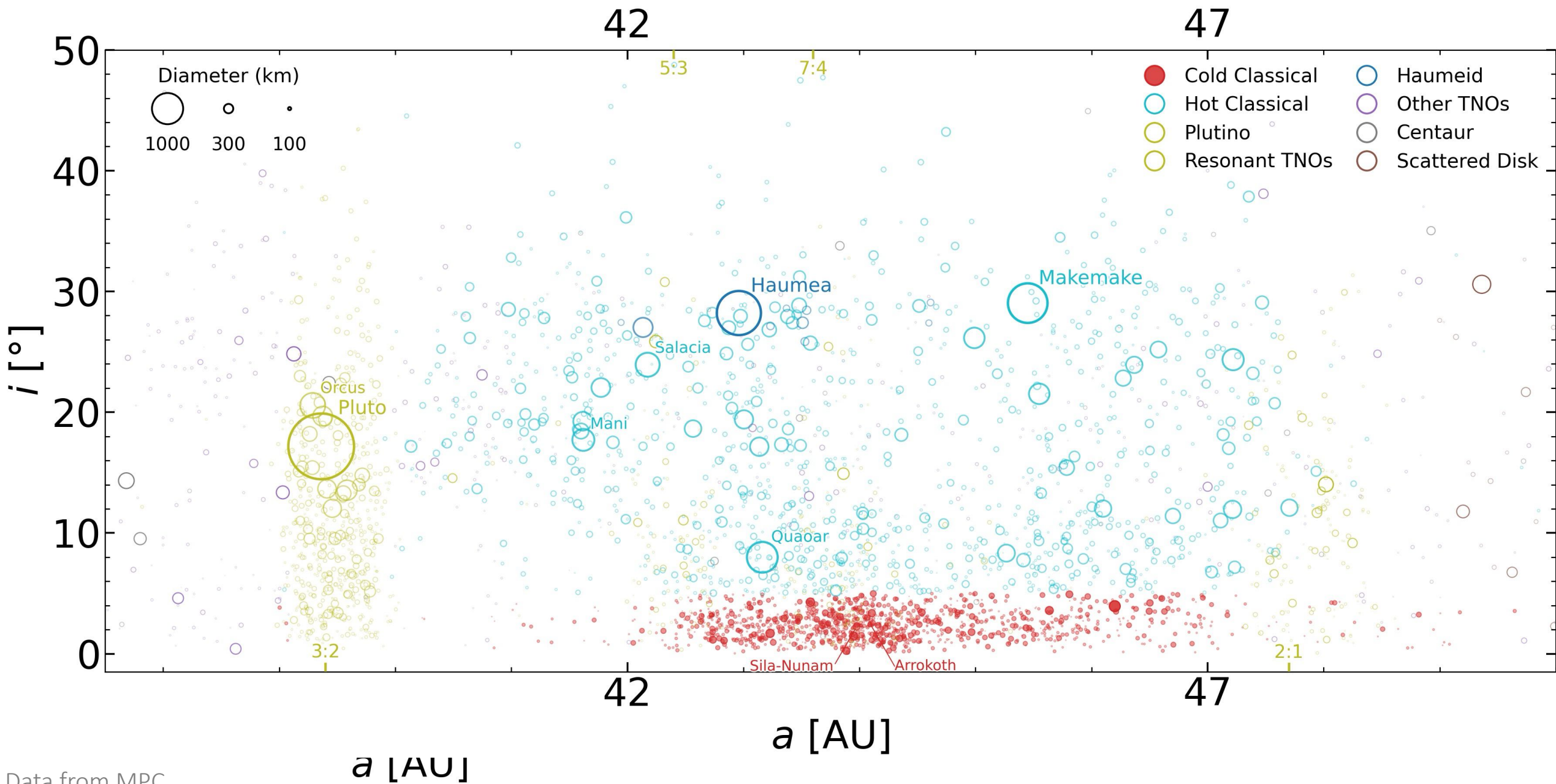
Data from the Minor Planet Center (MPC)

The Cold Classical Kuiper Belt (CCKB)

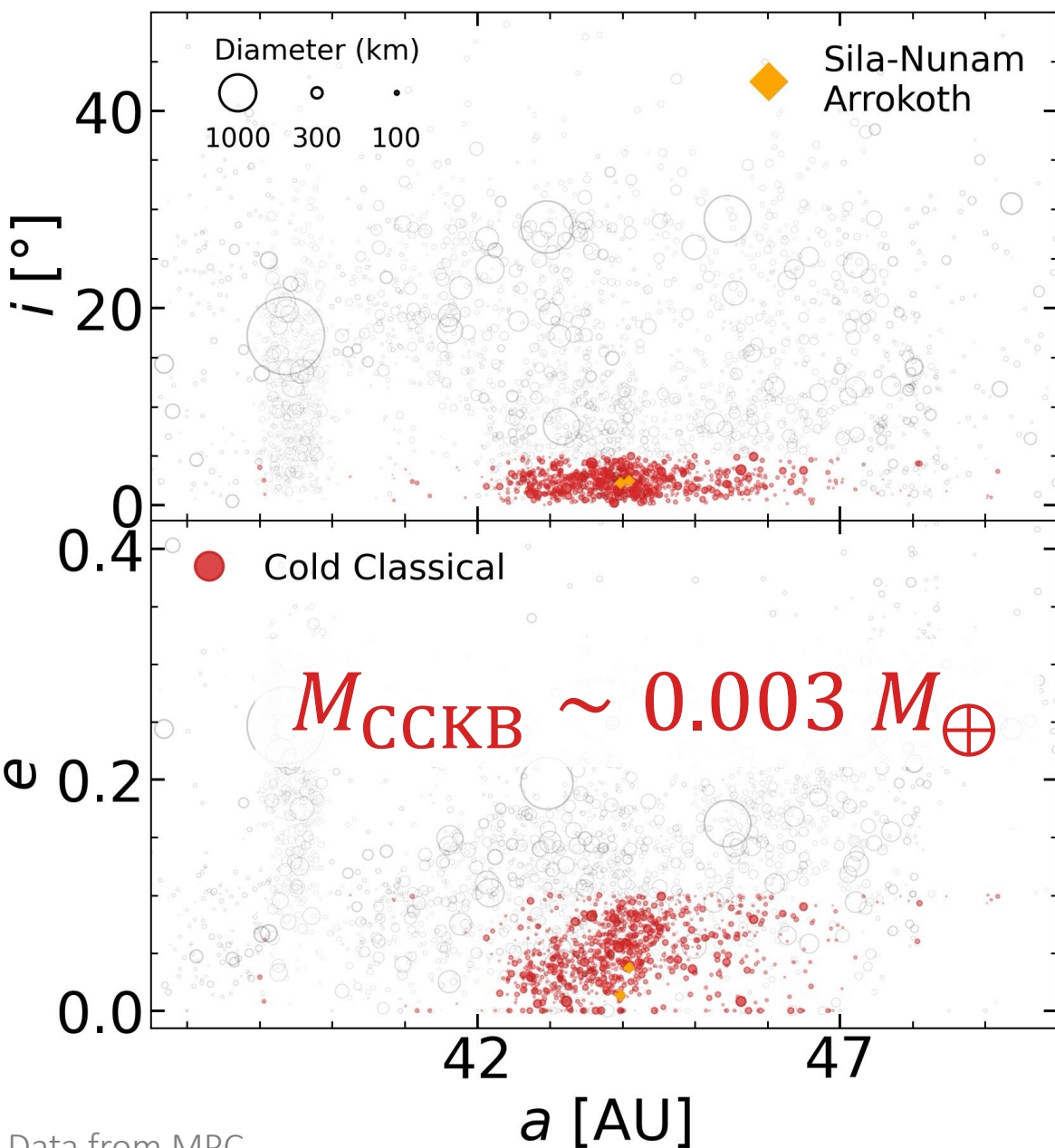


Data from the Minor Planet Center (MPC)

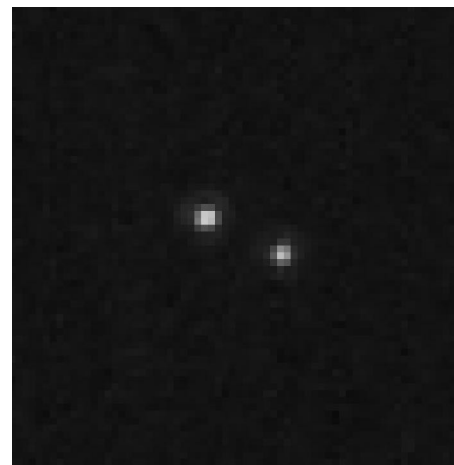
CCKBOs: Living Fossils of Planetesimals



CCKBOs: Living Fossils of *Binary* Planetesimals



Sila-Nunam

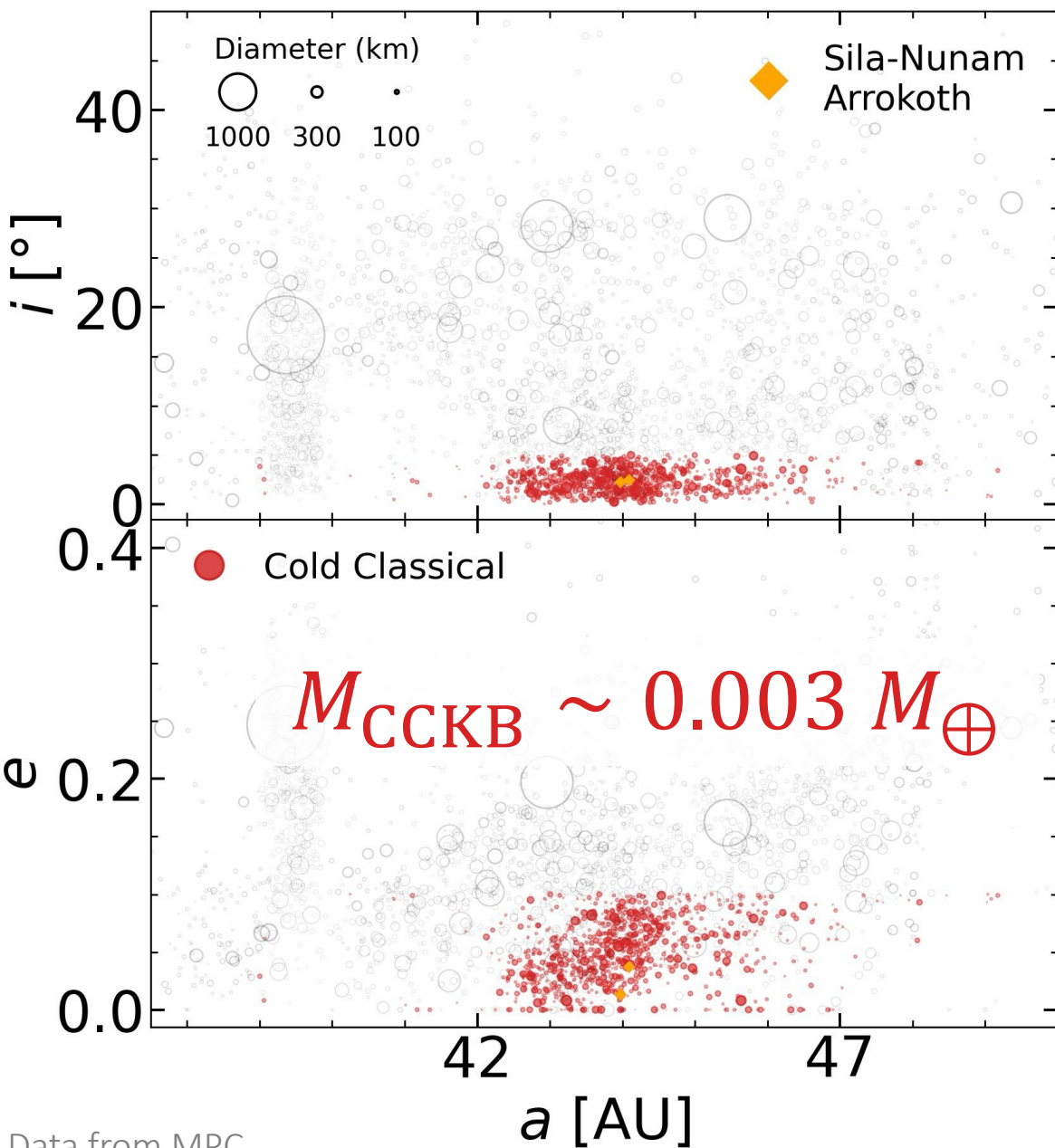


Arrokoth



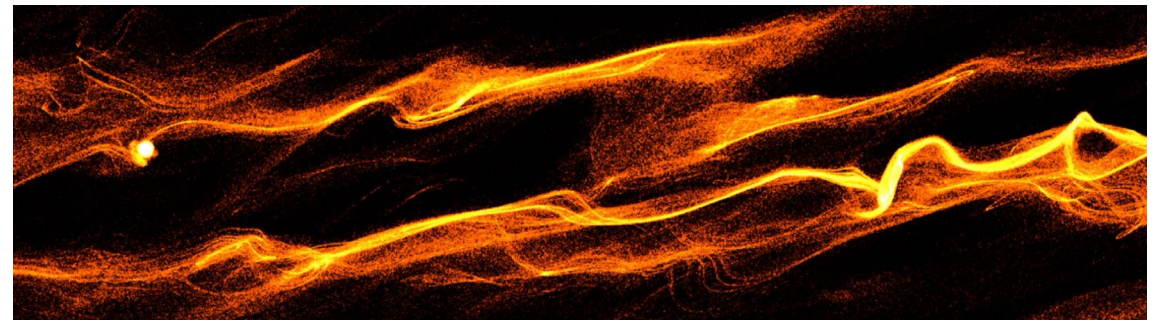
- a few hundred big objects
- “big”: ~ 100 km in size
- $\sim$ all binaries: 80% prograde, 20% retrograde

Forming the CCKB: Three Metrics



Data from MPC

The Streaming Instability



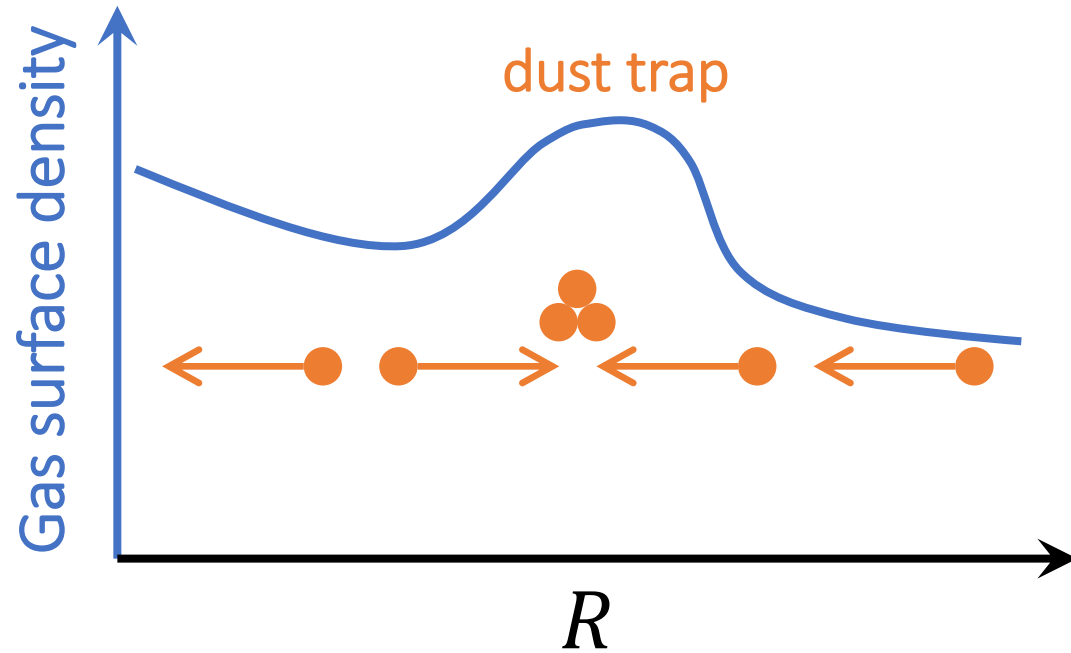
- Number (a few hundred)
- Size (~ 100 km)
- Binarity (80% vs 20%)

e.g., Youdin & Goodman 2005; Johansen+2007; Simon+2016, 2017; Schäfer+2017; Li+2019; Nesvorný+2019; Gerbig & Li 2023; etc.

In-Situ Formation of the Low-mass CCKB

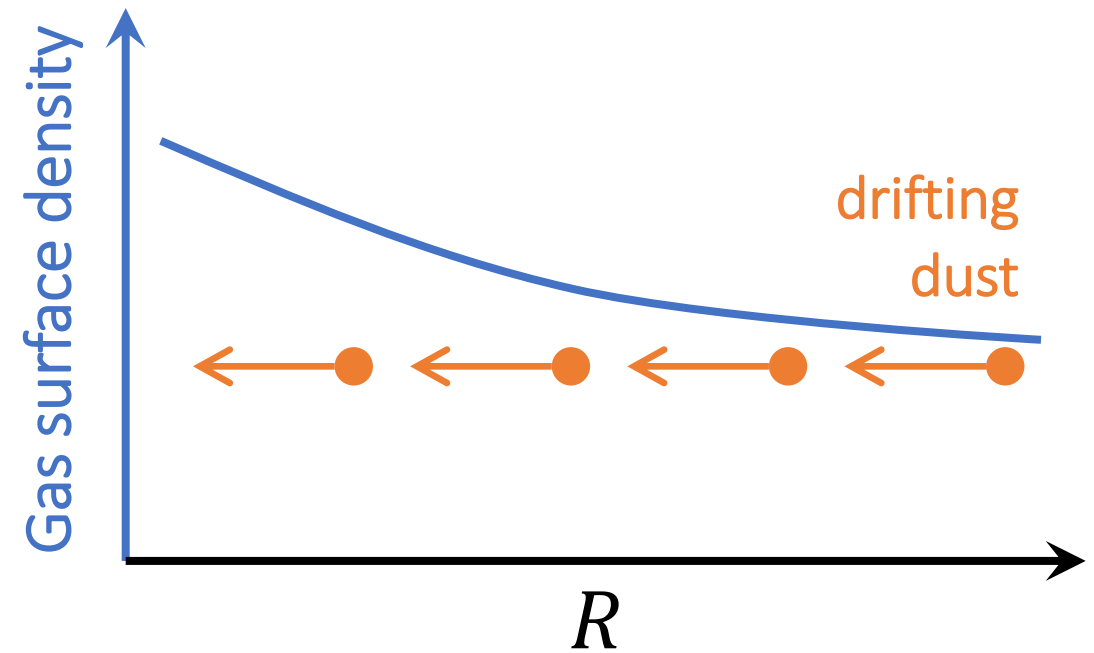
requires a late-stage, low-mass disk (a few % MMSN)

Scenario 1: Bump



- Number ?
- Size ?
- Binarity ?

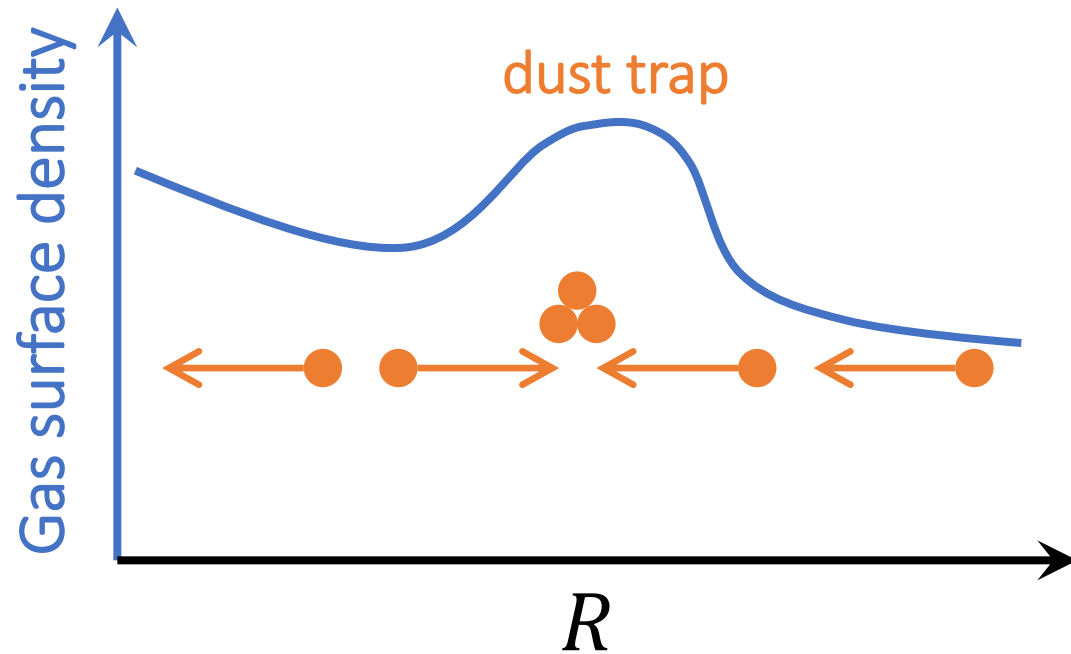
Scenario 2: No Bump



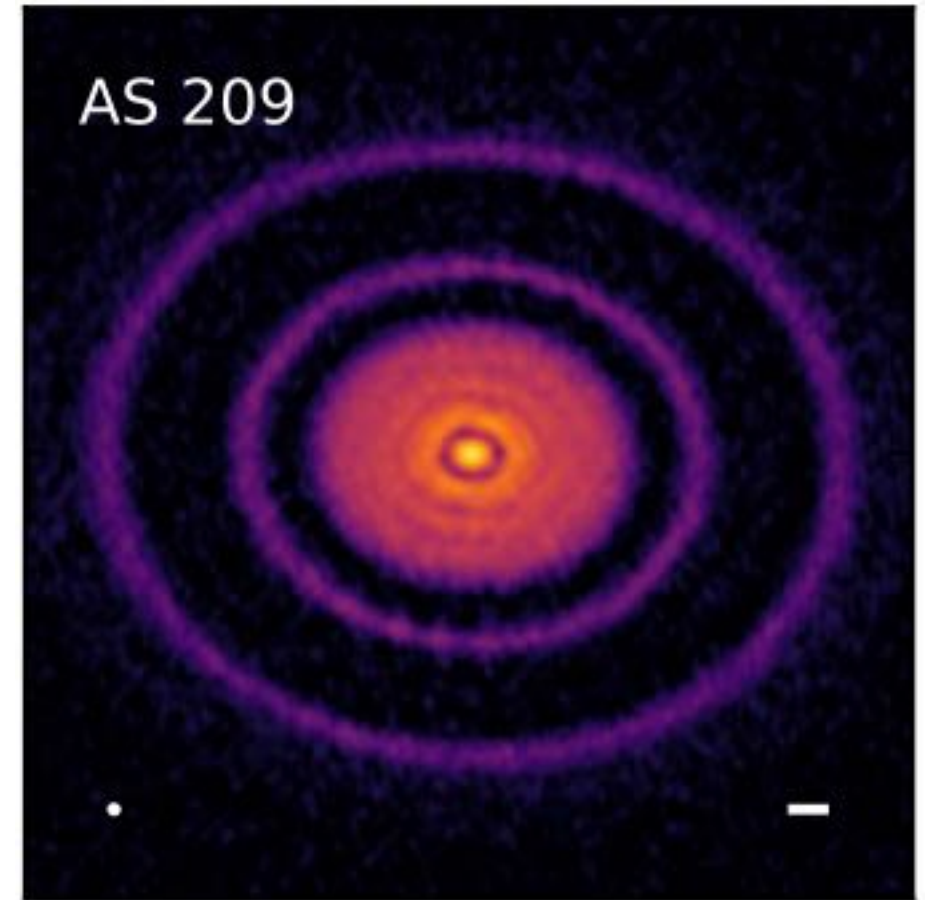
- Number ?
- Size ?
- Binarity ?

In-Situ Formation of the Low-mass CCKB in a Late-stage Disk

Scenario 1: Bump

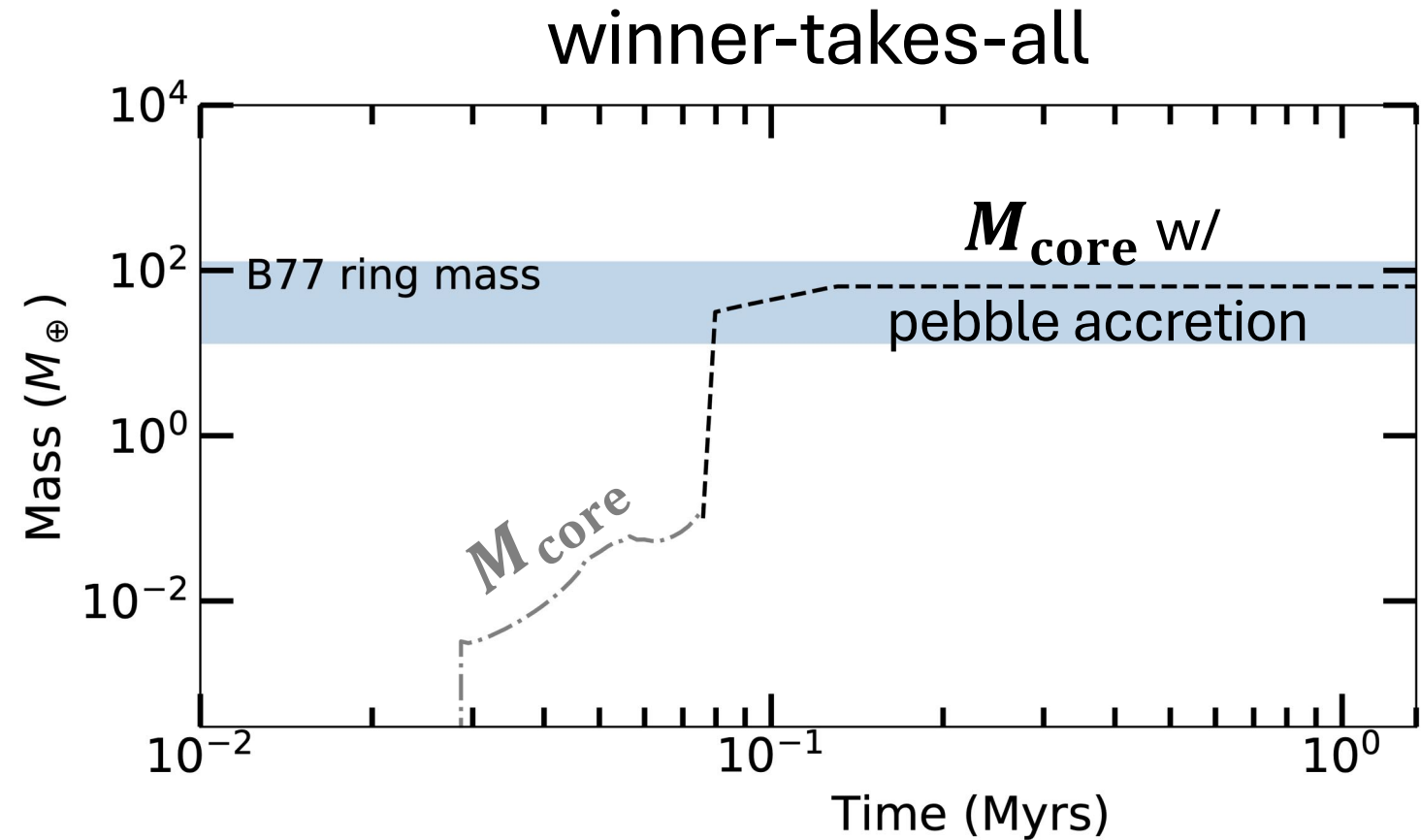
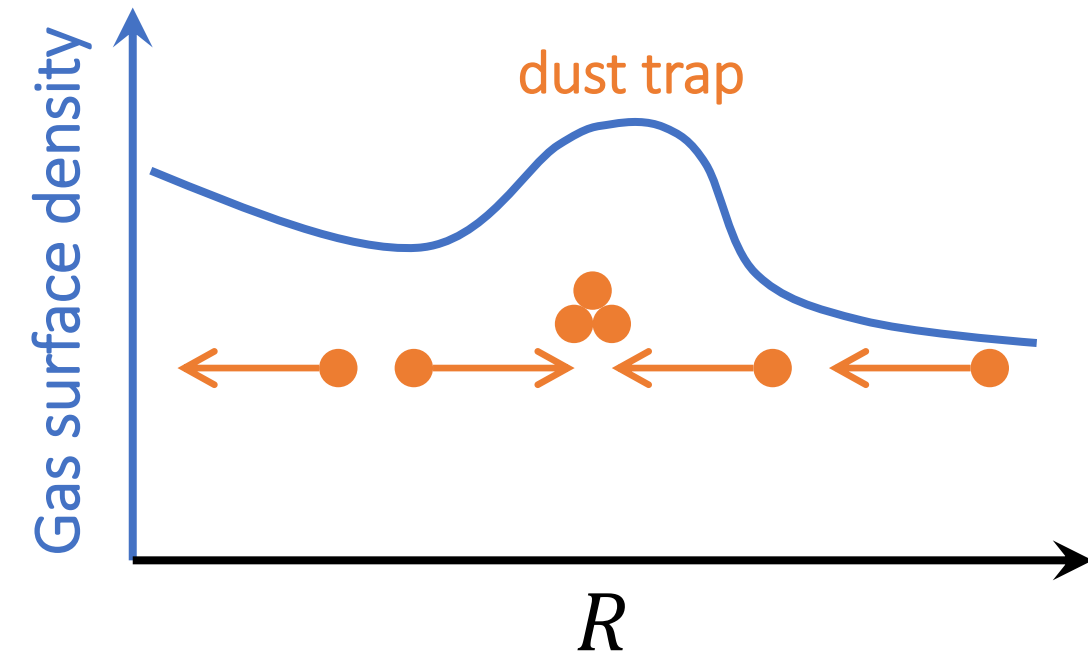


top-down view (dust continuum)



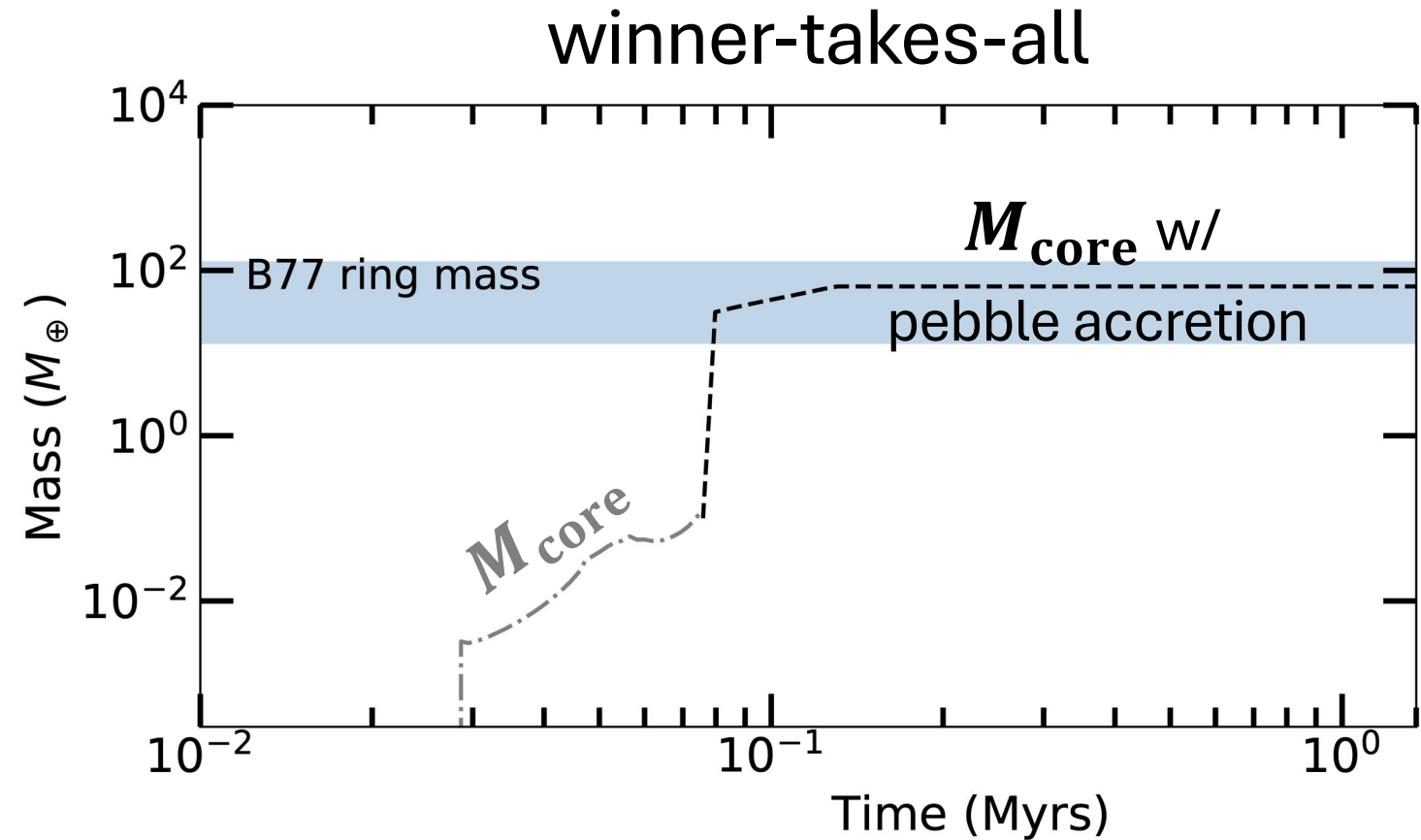
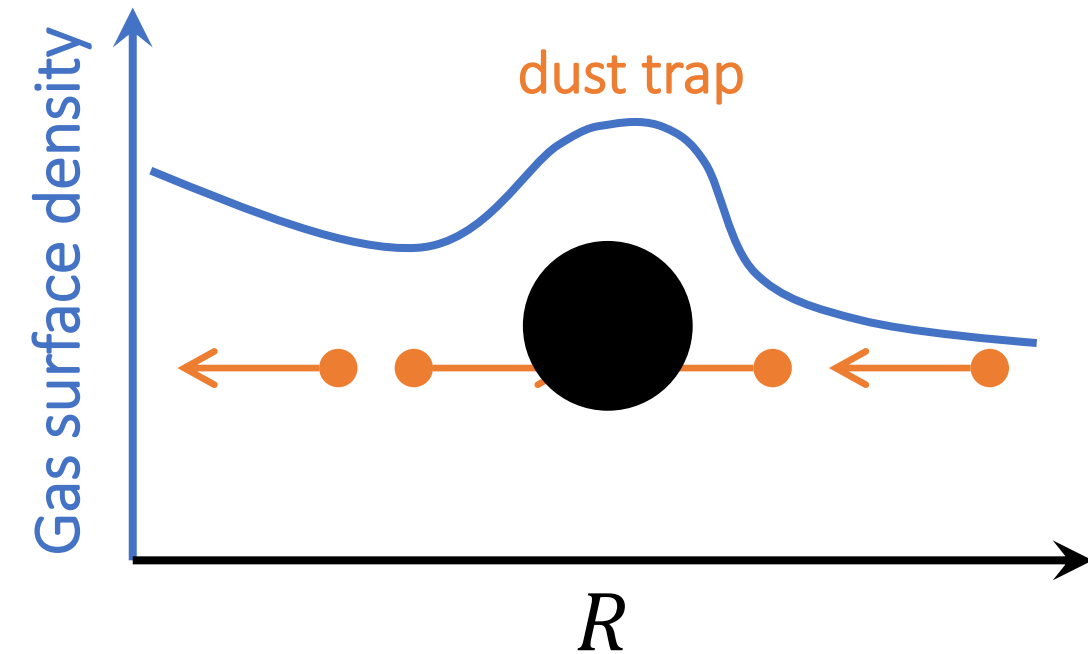
In-Situ Formation of the Low-mass CCKB in a Late-stage Disk

Scenario 1: Bump



In-Situ Formation of the Low-mass CCKB in a Late-stage Disk

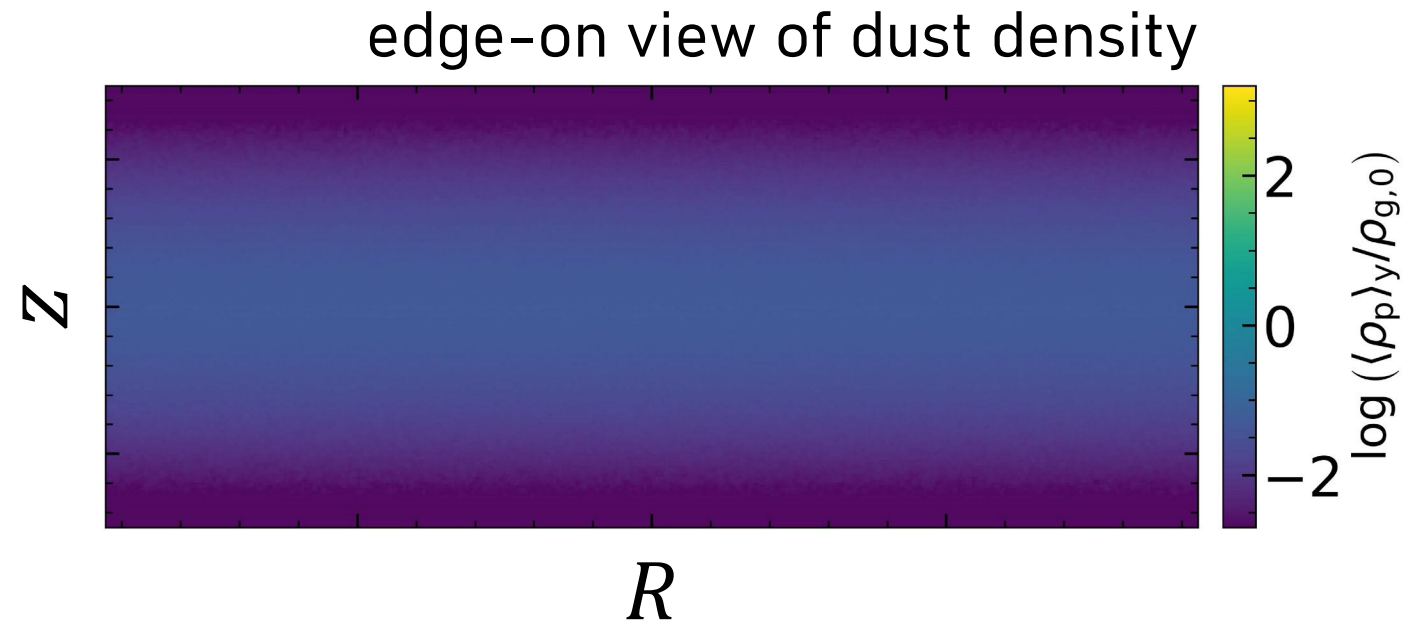
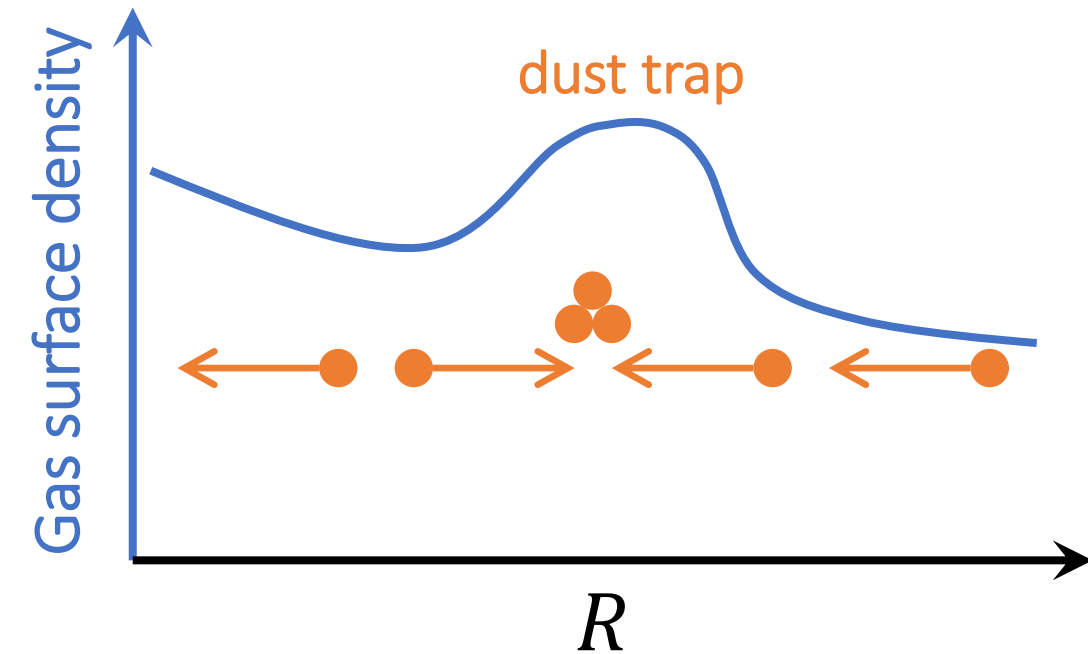
Scenario 1: Bump



In-Situ Formation of the Low-mass CCKB in a Late-stage Disk

- w/ forced turbulence ($\alpha \sim 10^{-4}$)
- $1 \times M_{\text{CCKB}}$ dust

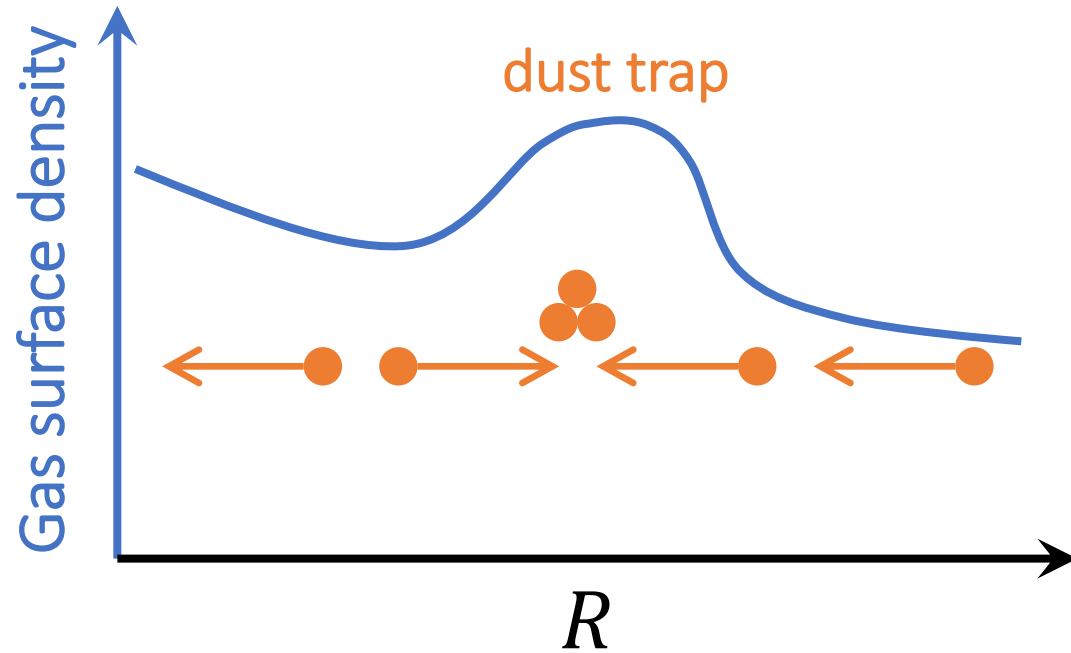
Scenario 1: Bump



In-Situ Formation of the Low-mass CCKB in a Late-stage Disk

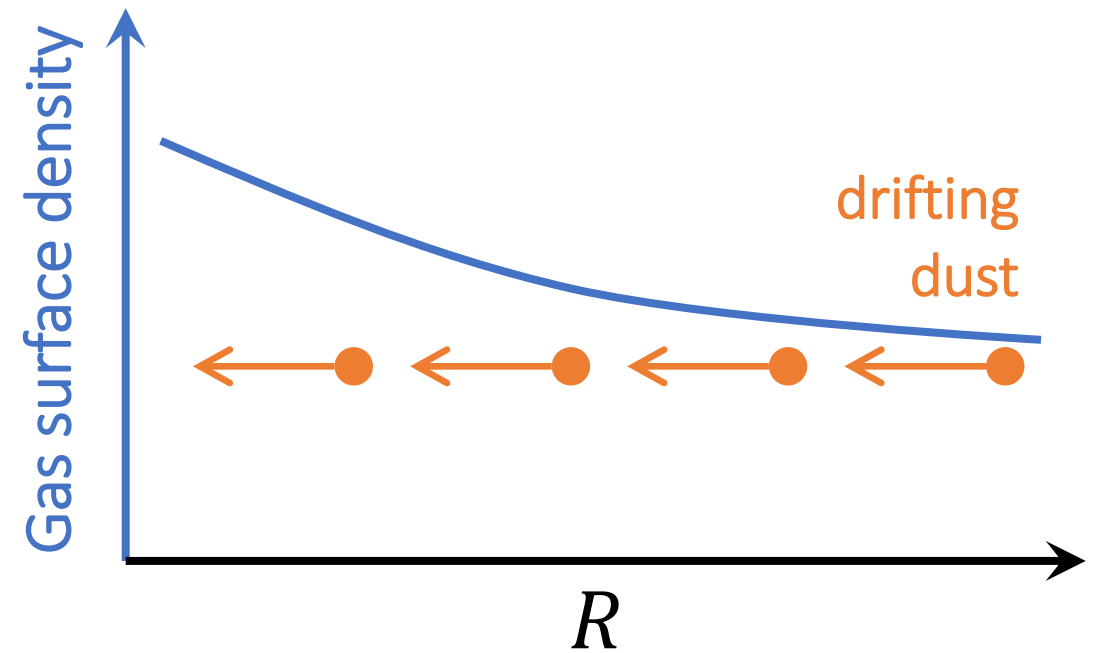
Unlikely

Scenario 1: Bump



- Number **X**
- Size **X**
- Binarity **X**

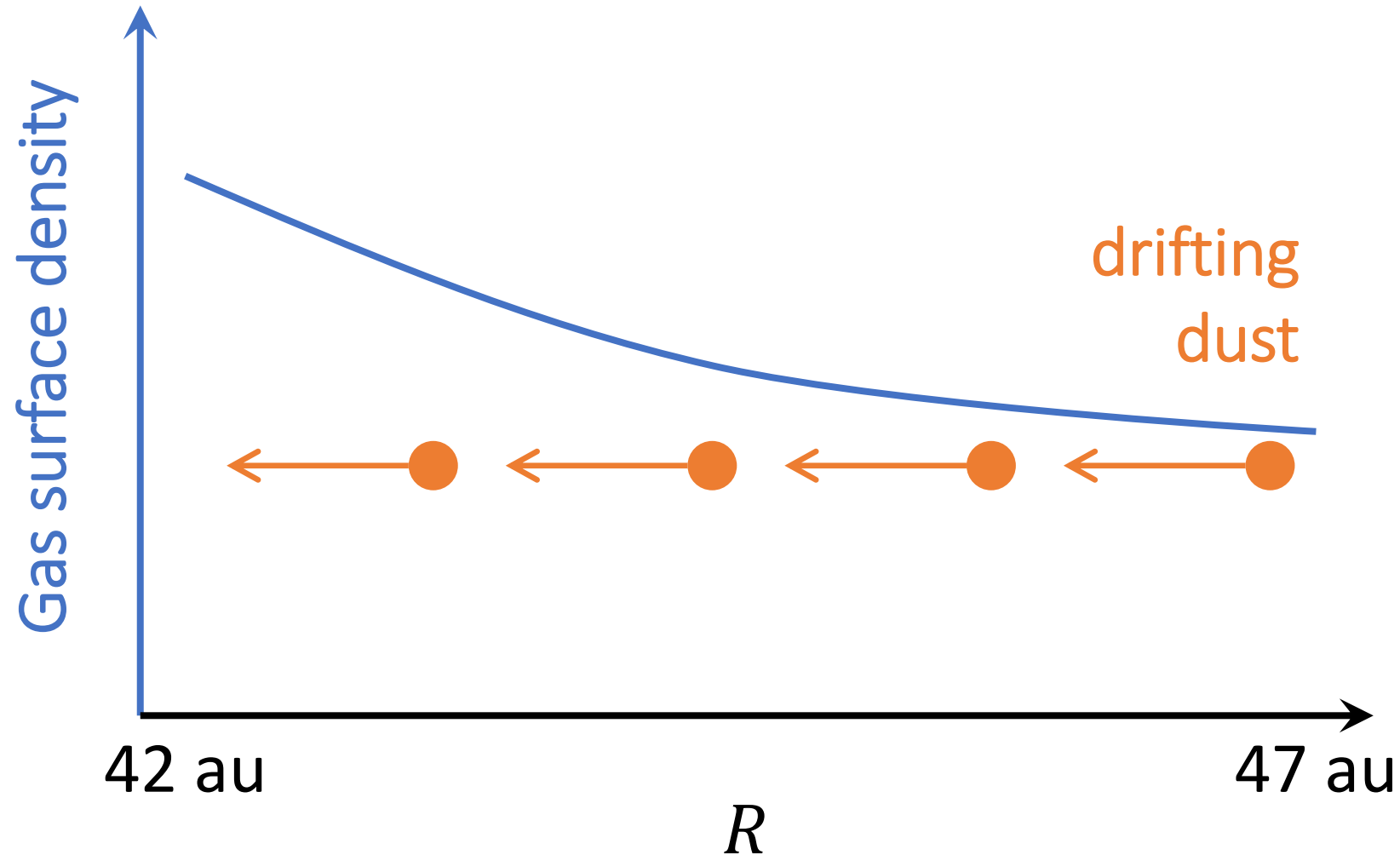
Scenario 2: No Bump



- Number ?
- Size ?
- Binarity ?

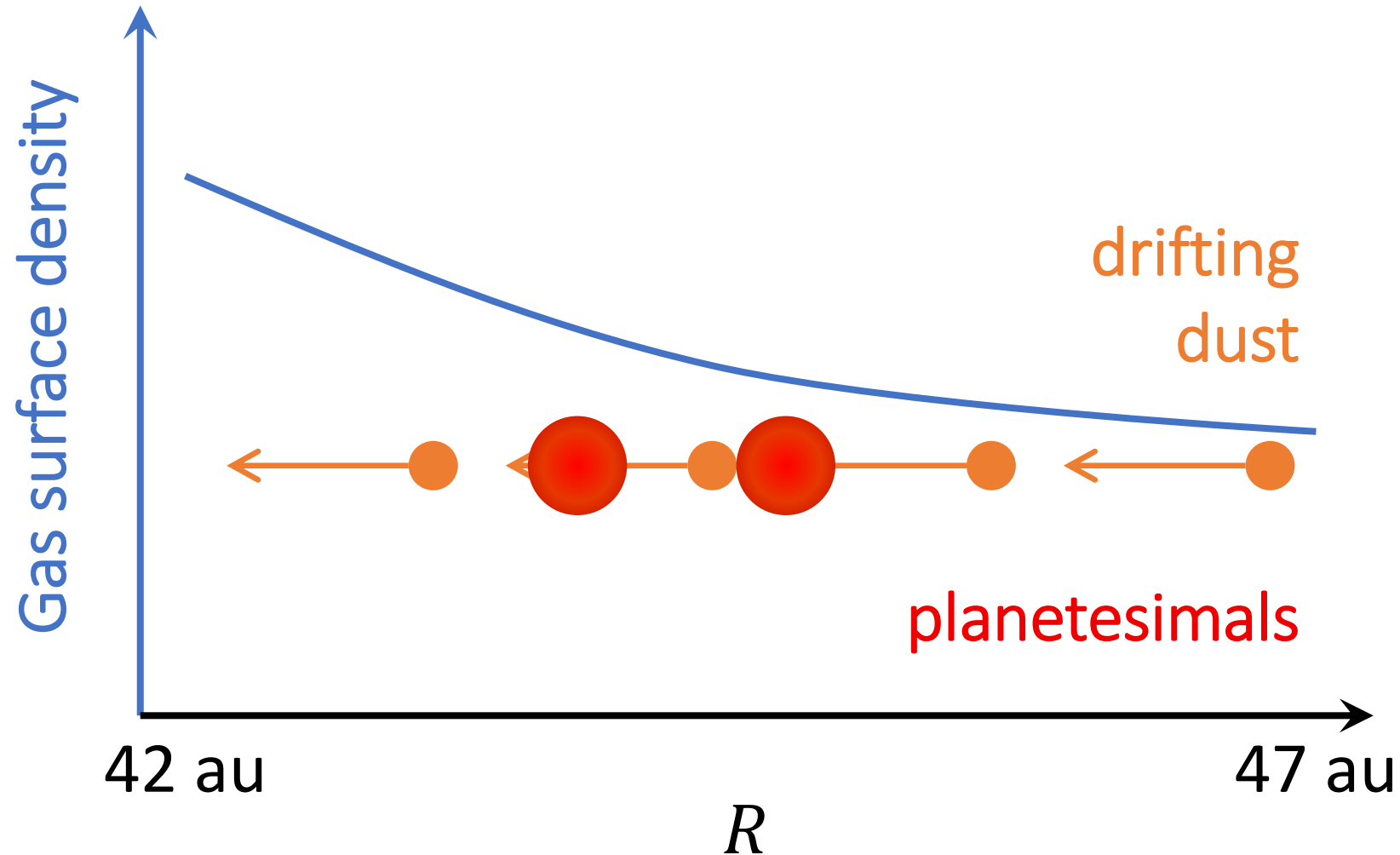
In-Situ Formation of the Low-mass CCKB in a Late-stage Disk

Scenario 2: No Bump



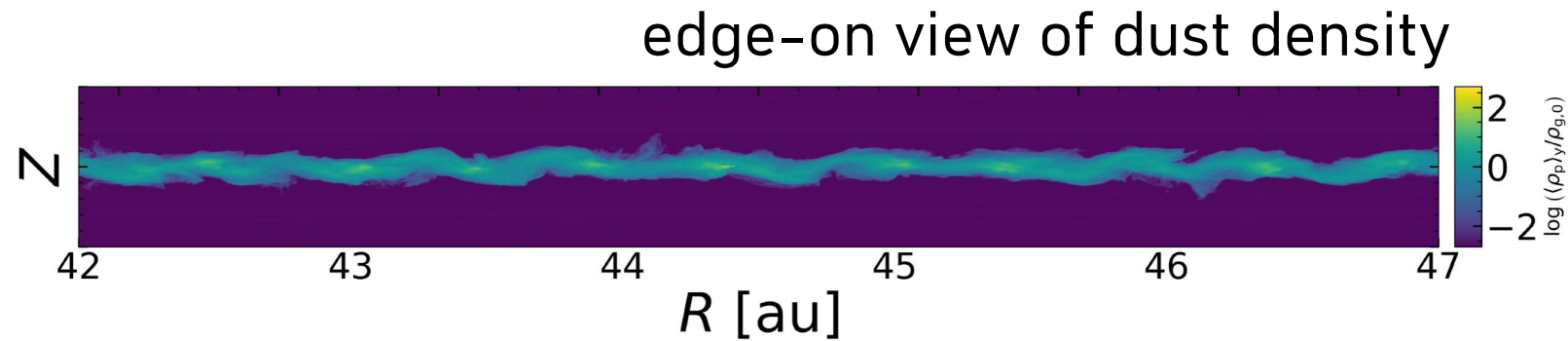
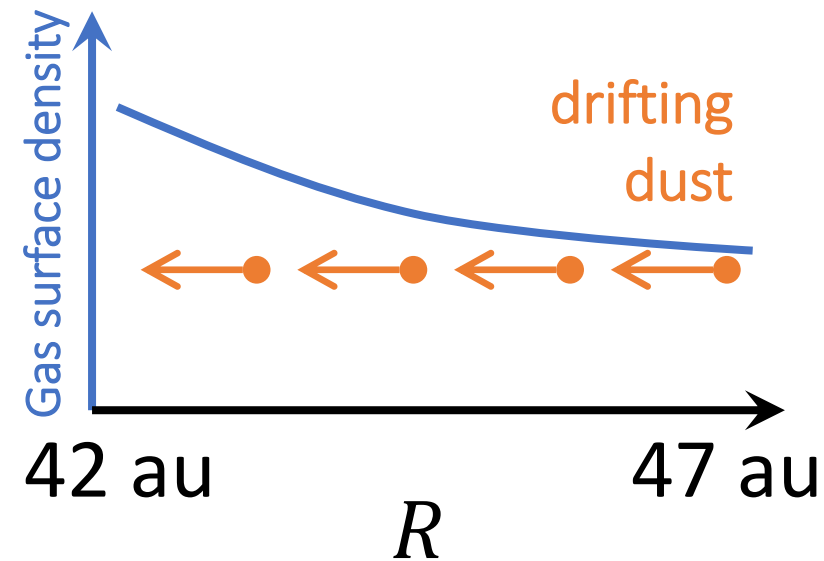
In-Situ Formation of the Low-mass CCKB in a Late-stage Disk

Scenario 2: No Bump



In-Situ Formation of the Low-mass CCKB in a Late-stage Disk

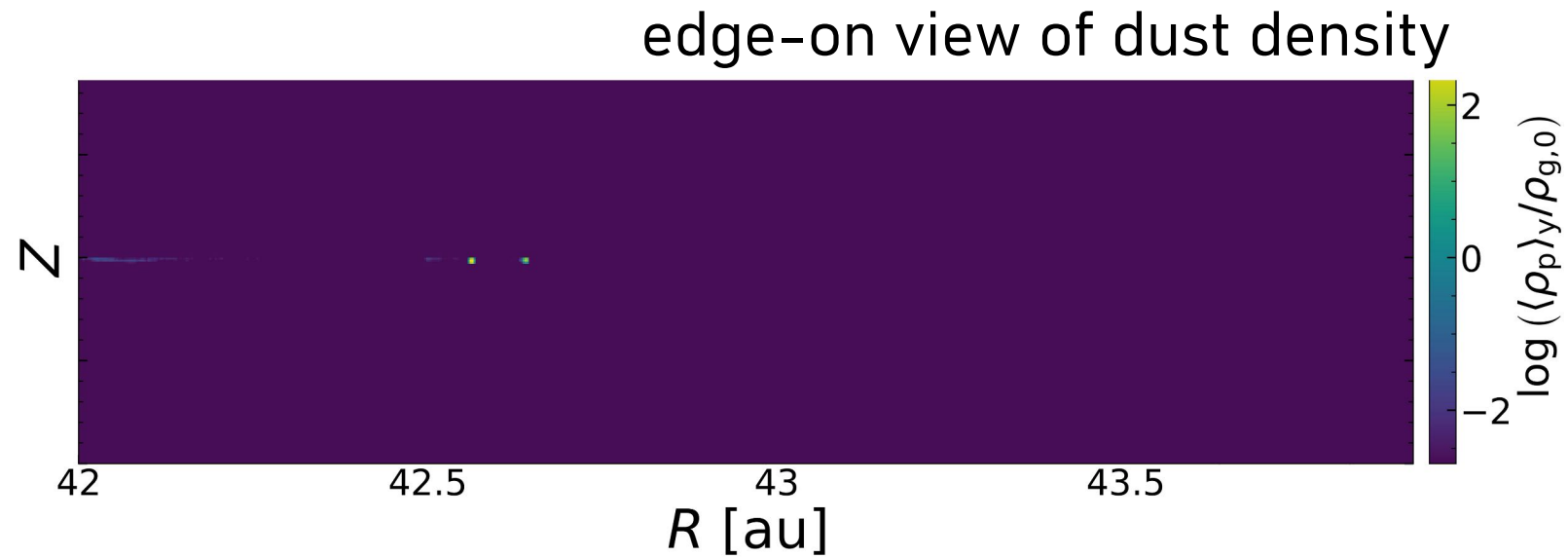
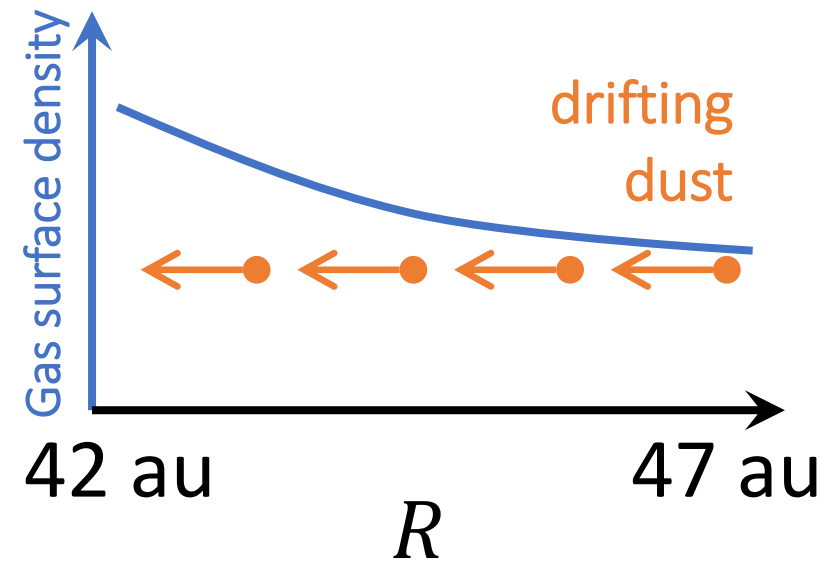
Scenario 2: No Bump



In-Situ Formation of the Low-mass CCKB in a Late-stage Disk

➤ reproduces size (~ 100 km)

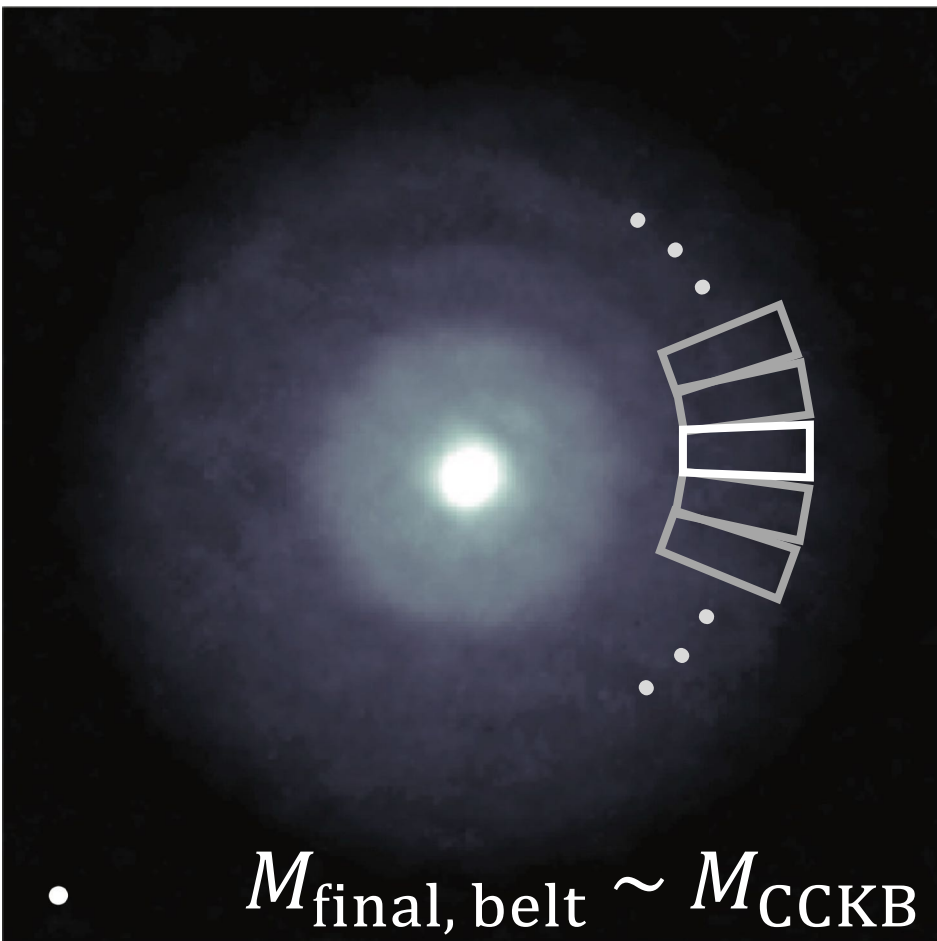
Scenario 2: No Bump



In-Situ Formation of the Low-mass CCKB in a Late-stage Disk

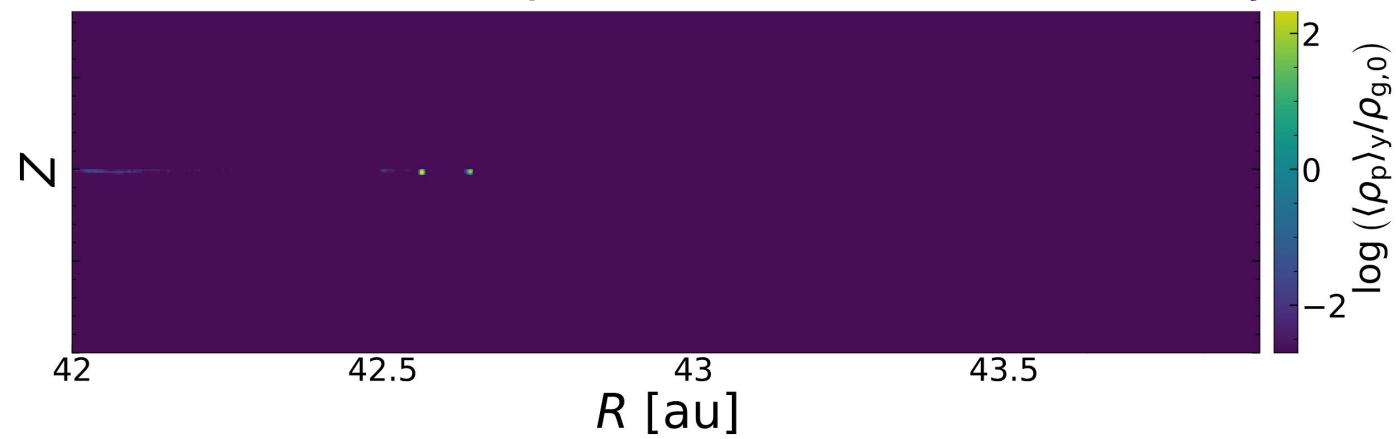
Scenario 2: No Bump

top-down view (gas disk)



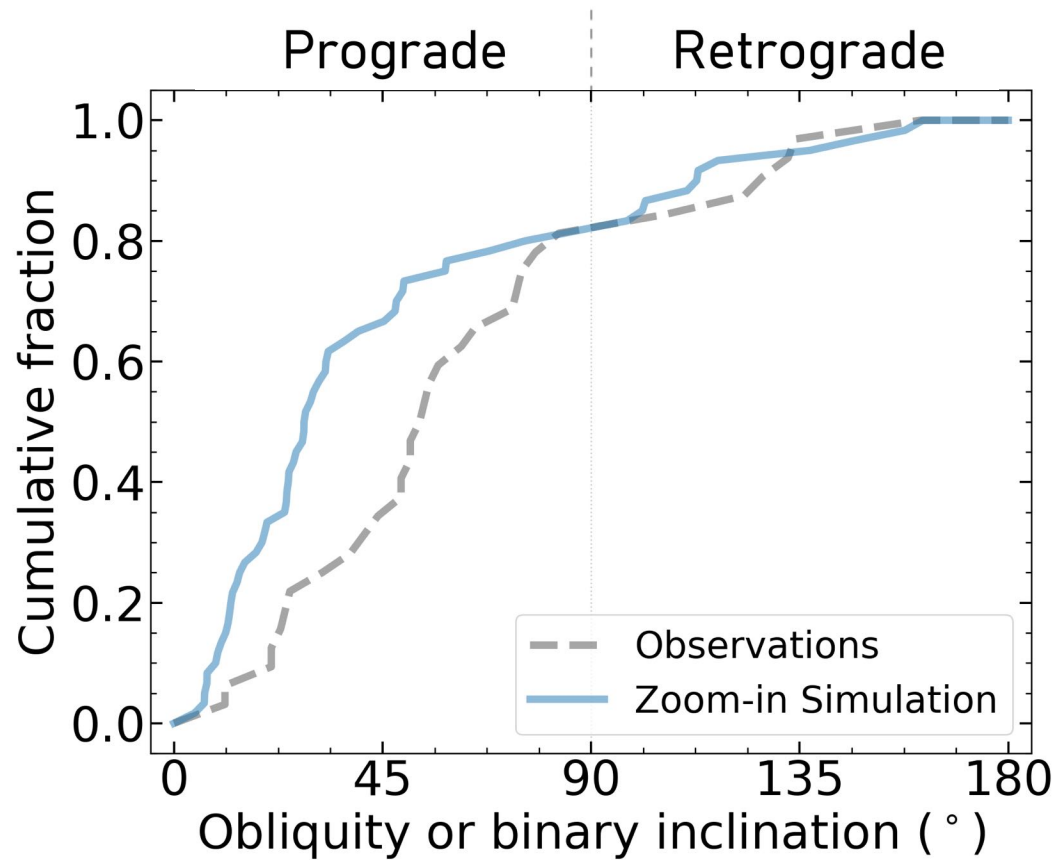
- reproduces size (~ 100 km)
- reproduces number (a few hundred)
=> reproduces the mass
- $50 \times M_{\text{CCKB}}$ dust initially, 98% drift away

edge-on view of dust density



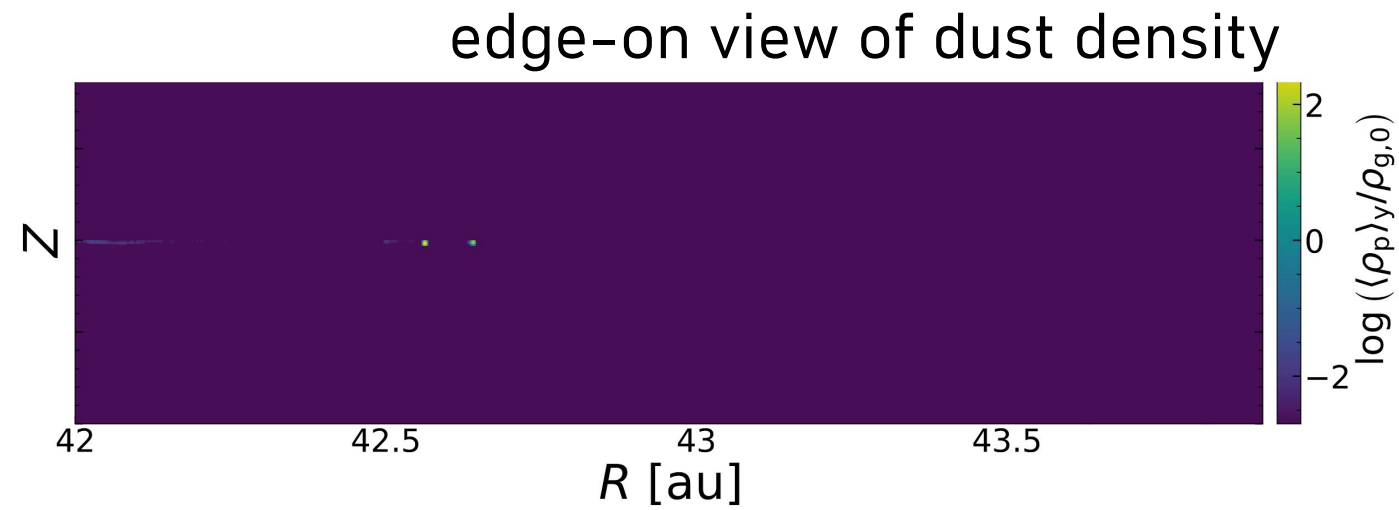
In-Situ Formation of the Low-mass CCKB in a Late-stage Disk

Scenario 2: No Bump



- reproduces size (~ 100 km)
- reproduces number (a few hundred)
=> reproduces the mass
- reproduces binarity (80% vs 20%)

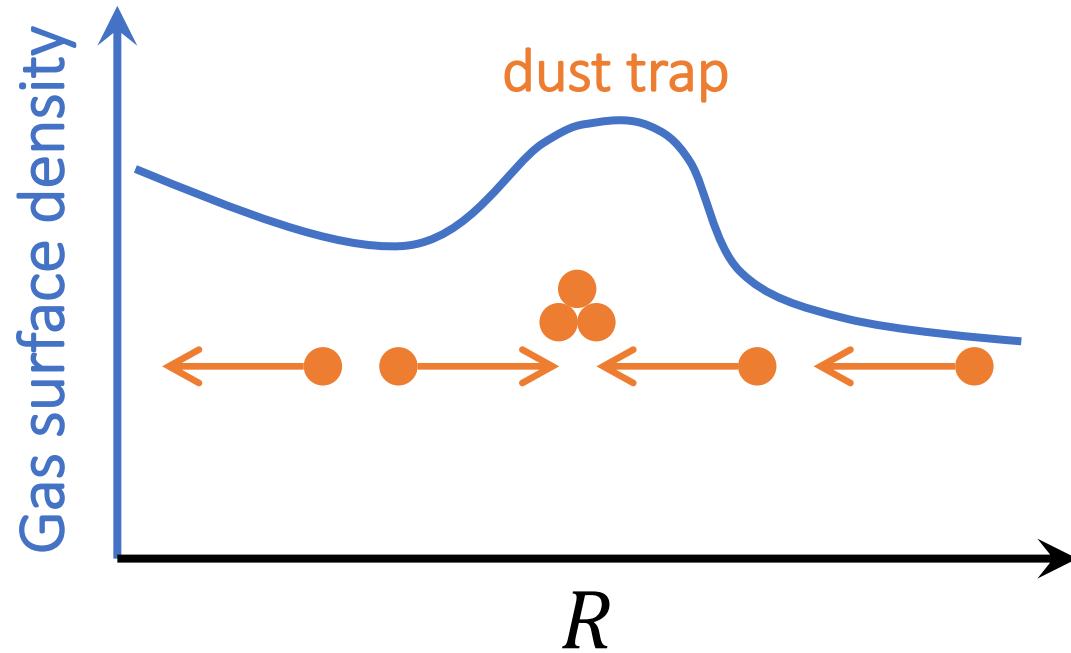
$$\frac{J_{\text{clump}}}{J_{\text{crit}}} = \frac{M_{\text{clump}} \cdot \eta v_K \cdot \eta r}{J_{\text{crit}}} \gg 1$$



In-Situ Formation of the Low-mass CCKB in a Late-stage Disk

Unlikely

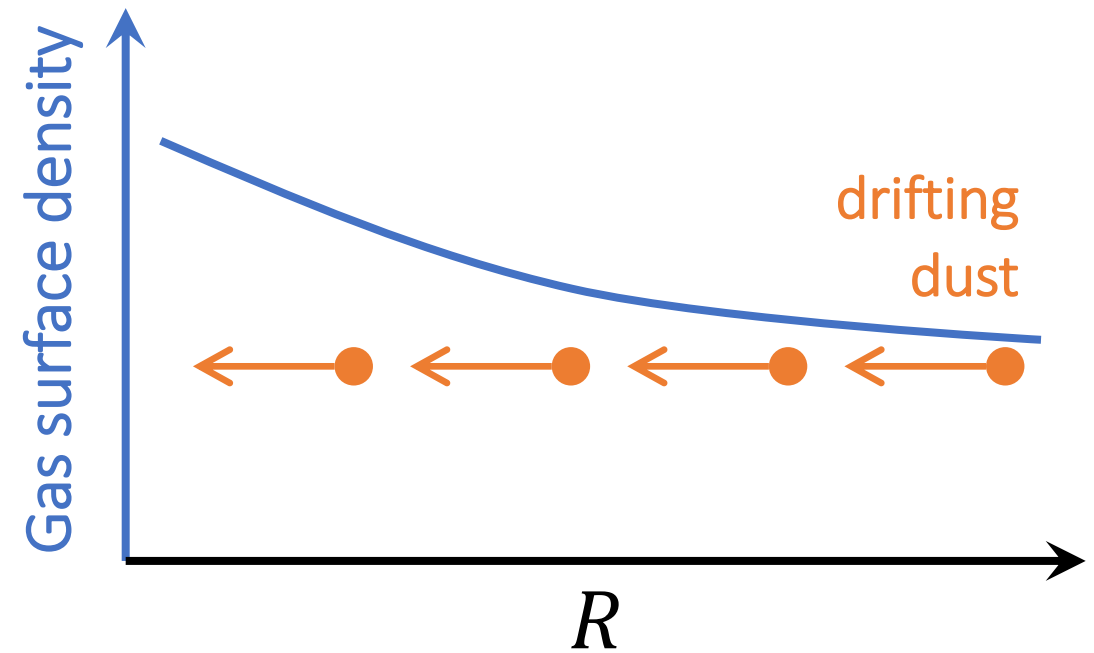
Scenario 1: Bump



- Number **X**
- Size **X**
- Binarity **X**

Favored

Scenario 2: No Bump

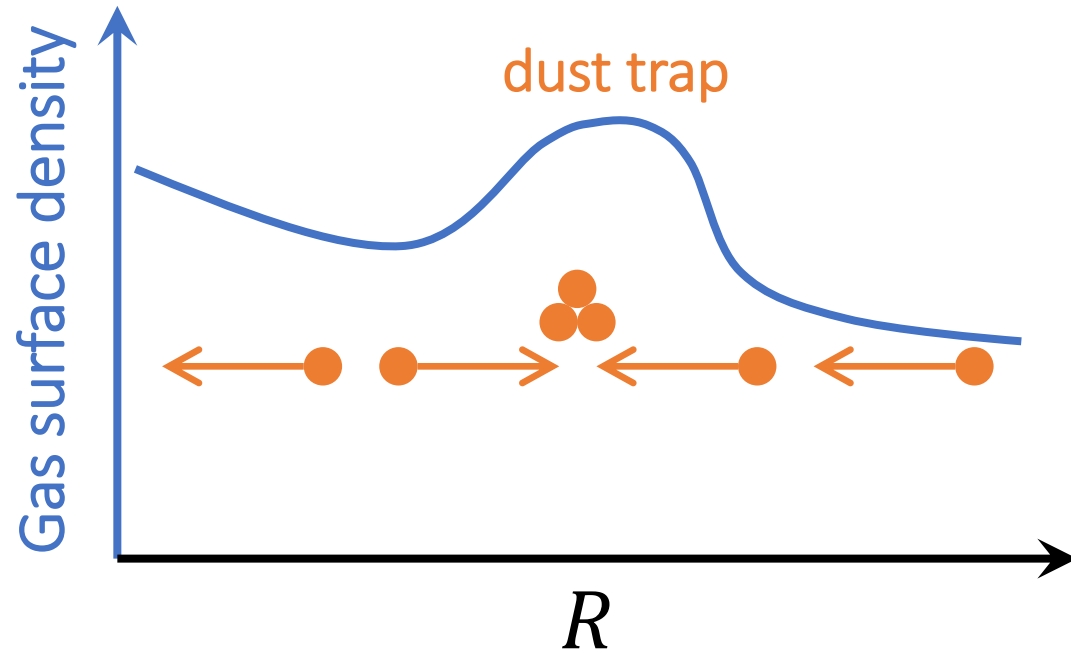


- Number ✓
- Size ✓
- Binarity ✓

In-Situ Formation of the Low-mass CCKB in a Late-stage Disk

Unlikely

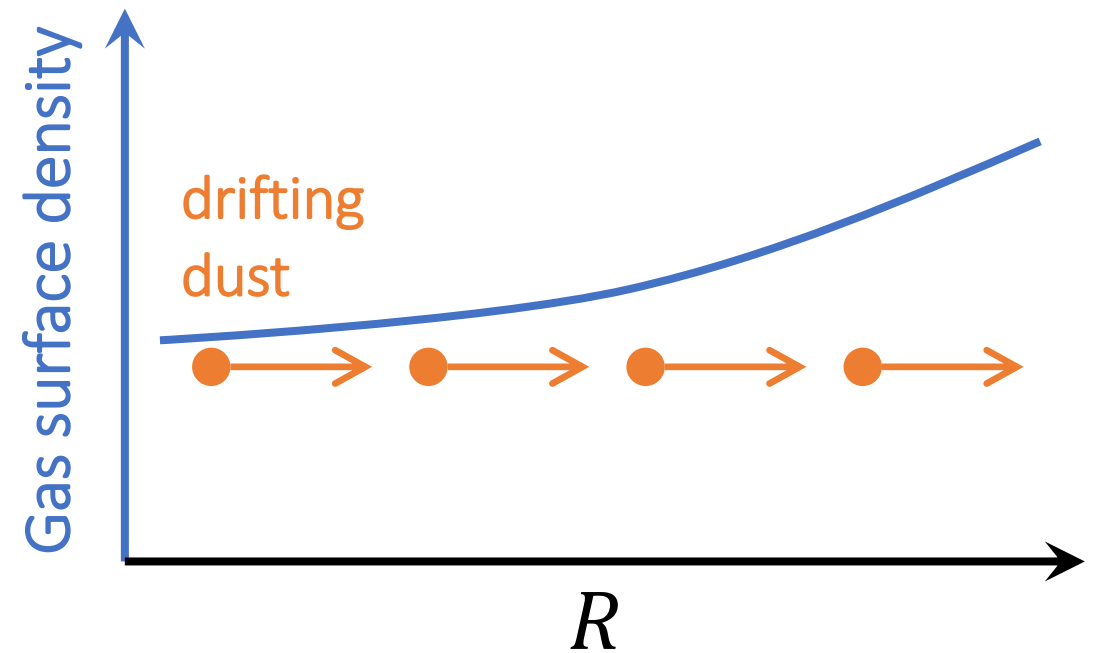
Scenario 1: Bump



- Number **X**
- Size **X**
- Binarity **X**

Favored

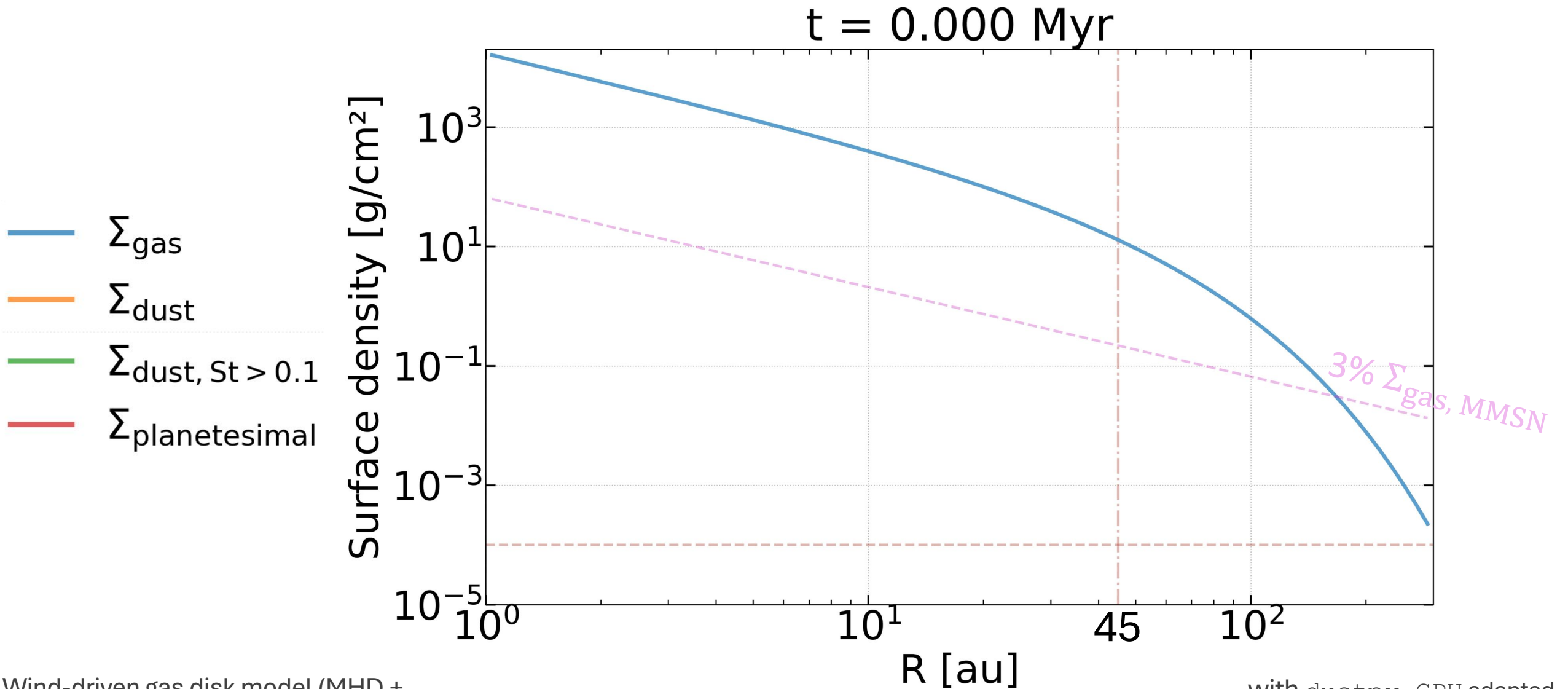
Scenario 2: No Bump



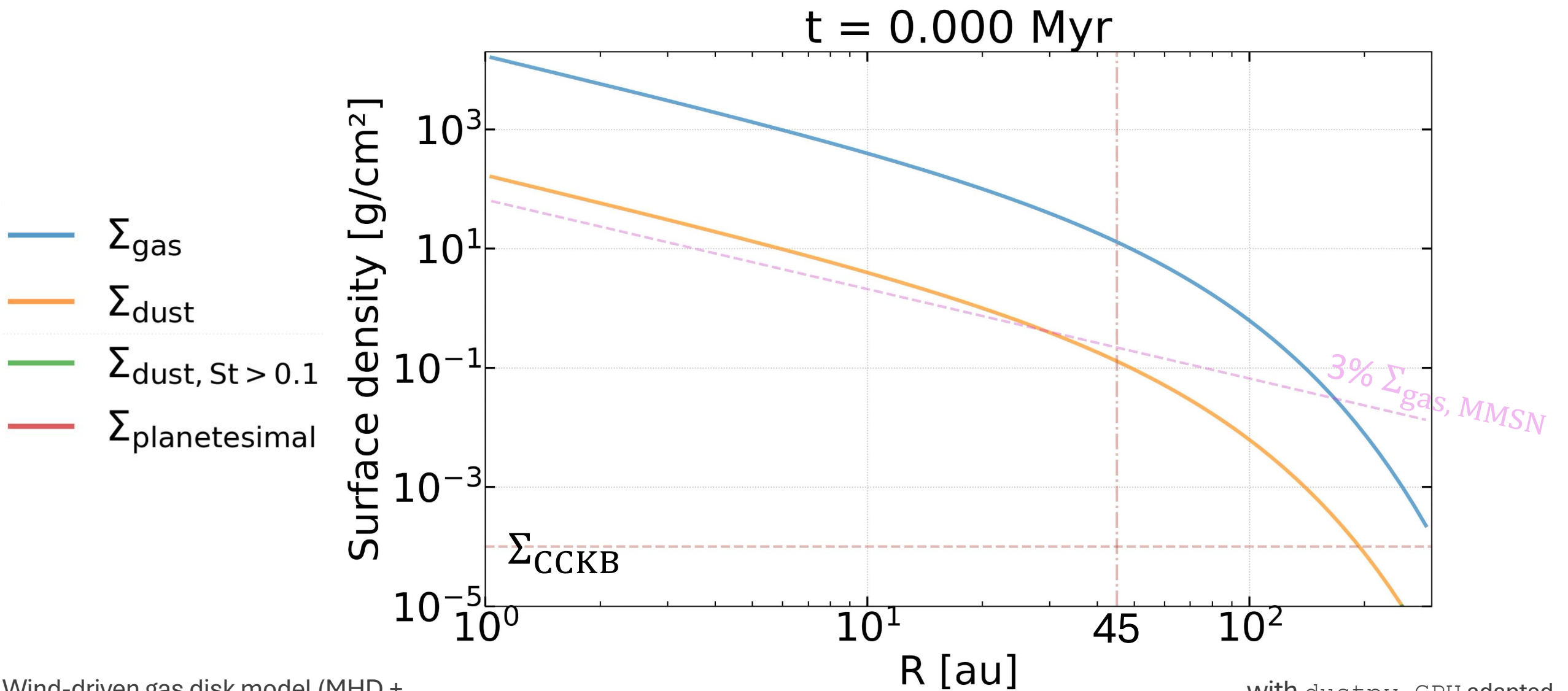
- Number ✓
- Size ✓
- Binarity ✓

Forming the CCKB — What's the Backstory?

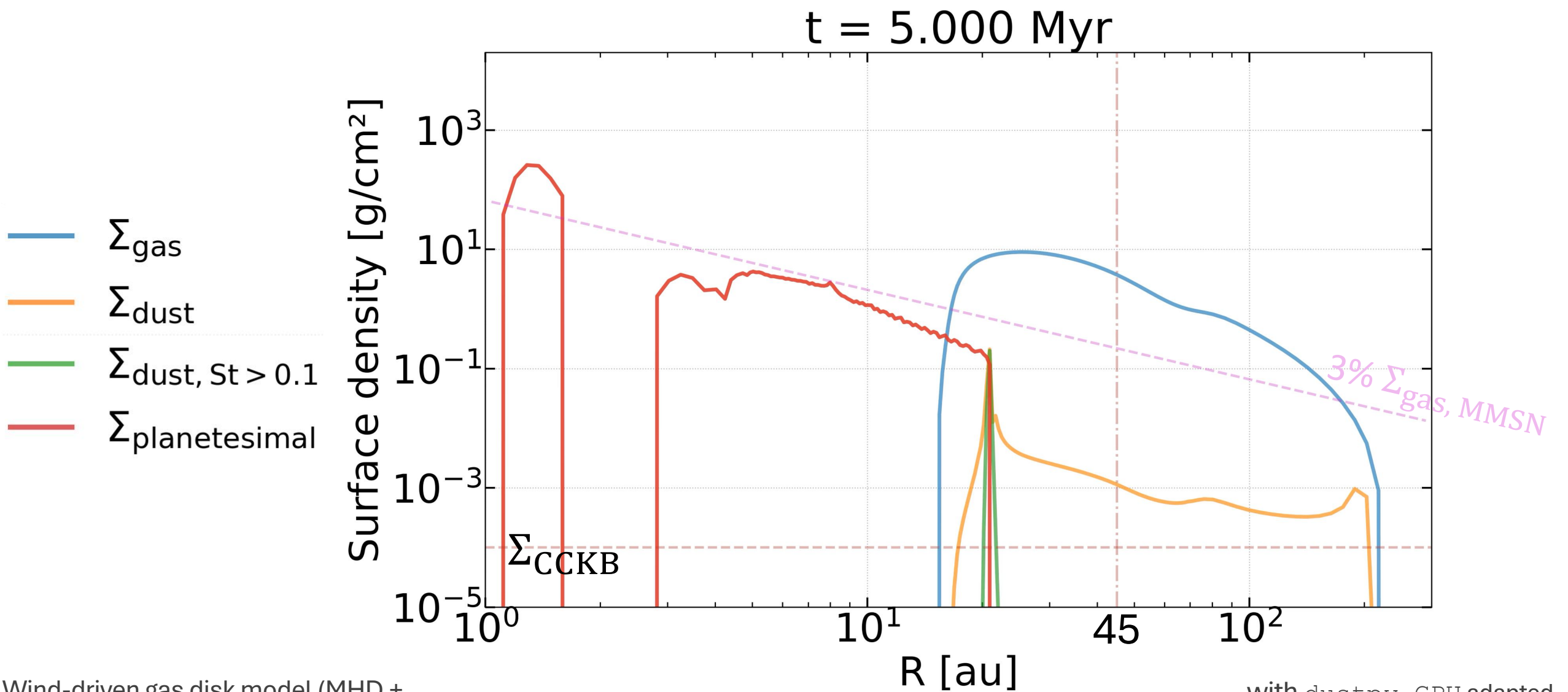
Wind-driven dissipation triggers the last wave of planetesimal formation?



Forming the CCKB — What's the Backstory?

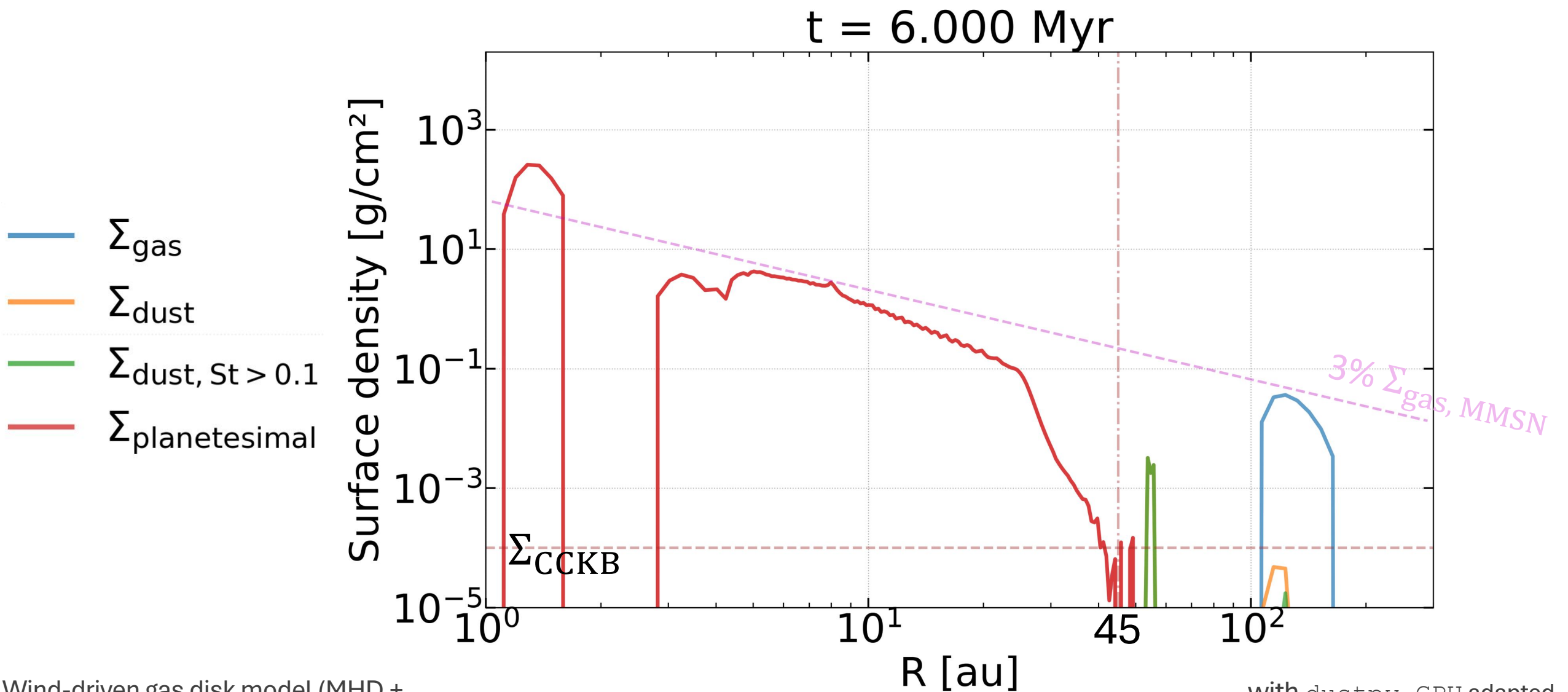


Forming the CCKB — What's the Backstory?



An Origin Story for the CCKB and the Kuiper Cliff

Wind-driven inside-out dissipation paves a planetesimal disk that falls off at the Kuiper Cliff



Wind-driven gas disk model (MHD +
photoevaporative) from Kunitomo+2020

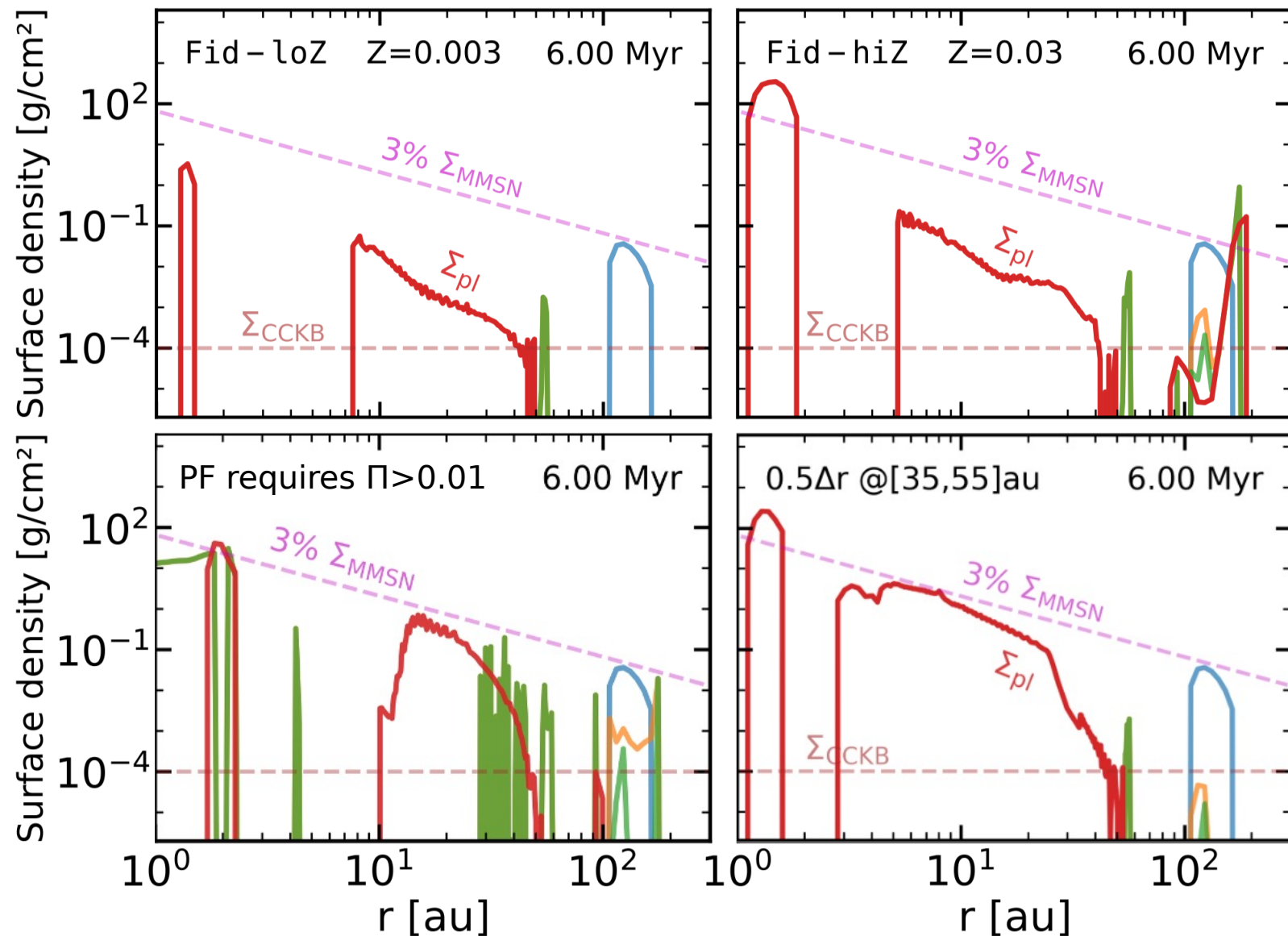
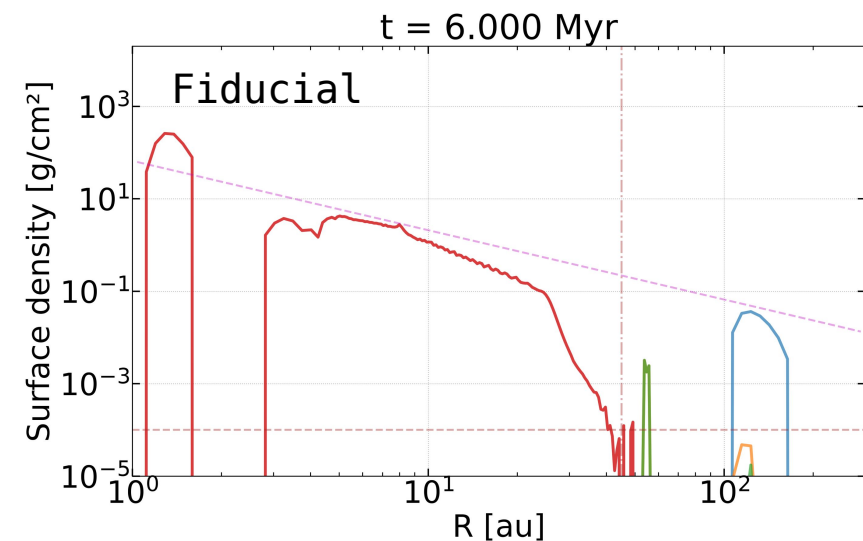
Li & Chiang, on arXiv soon

with dustpy-GPU adapted
from Stammer & Birnstiel 2022

An Origin Story for the CCKB and the Kuiper Cliff

Wind-driven inside-out dissipation paves a planetesimal disk that falls off at the Kuiper Cliff

And this feature is insensitive to a variety of factors:



Summary

- Streaming Instability is a robust & efficient mechanism for planetesimal formation
- Most planetesimals form as binaries
 - an expected outcome of the SI
- CCKB formed late in the dissipating solar nebula
 - “drift-and-form” scenario can reproduce the total mass in CCKBOs, as well as their characteristic size and binarity
- a turbulent gas pressure bump does not help
- *an origin story*: a generic gas disk dissipates inside-out reproduces the Kuiper Cliff and Σ_{CCKB}

